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SieveComP: Automated and standardized seismic catalog processing from SeisComP

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Software

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Summary

SeisComP is a comprehensive seismological software platform designed for real-time data acquisition, automatic processing, and interactive event analysis. Many seismic monitoring networks worldwide rely heavily on it as the core of their operational processing systems. However, although the initial automatic event detection and location are complete, performing routine quality control, identifying duplicate events, and standardizing the resulting seismic catalogs often requires significant manual effort from analysts or the continuous maintenance of ad-hoc scripts.

SieveComP is an open-source Python package designed to automate, standardize, and accelerate the post-processing and revision of seismic catalogs generated by SeisComP. It is built to accommodate diverse technical backgrounds: beginner seismologists and network operators can leverage human-readable, declarative TOML configuration files to define complex validation rules (such as geographic bounding boxes, custom polygonal zones, depth constraints, and duplicate event detection) in just a few lines of code. Simultaneously, SieveComP provides a robust, low-level Python API that grants advanced users and researchers a way to customize, extend, and directly manipulate the validation pipelines.

Furthermore, as the use of agentic Artificial Intelligence (AI) increases in scientific workflows, SieveComP serves as a crucial, secure abstraction layer. It provides a structured framework that allows AI-driven agents to independently monitor data quality, generate visualizations, design chatbots and automate routine reporting. By restricting AI interactions to SieveComP's well-defined engine, observatories can harness the efficiency of autonomous agents while mitigating the severe risks of unconstrained AI database manipulation (i.e., granting direct access to sensitive production databases).

In this Python package, SieveComP's core engine translates these rules into highly efficient, vectorized operations capable of processing thousands of events in seconds. It also supports database queries between different SeisComP versions and systems to provide a unified catalog view, making it a suitable tool to rapidly filter, audit, and curate high-quality seismic data.

Statement of Need

Ensuring the long-term quality of seismic data and catalogs from large networks remains a critical challenge in seismology, especially as the volume of available events continues to grow exponentially. While platforms like SeisComP (Hanka et al., 2010) are standard for initial real-time event detection and location, refining the resulting catalogs into a highly reliable dataset is historically a manual, non-standardized process disposed to human error and scaling bottlenecks.

Recent literature remarks the growing necessity for automated quality assurance frameworks

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across seismological workflows. For example, the *Scalable Seismological Pipeline for Assessment, Review, and Quality (SSPARQ)* was recently introduced to automate the assessment of station performance, orientation trends, and waveform quality (Coelho et al., 2026). Furthermore, foundational ecosystem tools like *ObsPy* provide extensive libraries for general seismological data analysis and waveform manipulation (Megies et al., 2011). However, there remains a gap in the software ecosystem for tools designed specifically to audit the downstream parametric catalogs produced by SeisComP databases. Network operators currently lack robust, out-of-the-box engines focused on the rule-based validation of magnitudes, depth thresholds, custom tectonic spatial distributions, and logical deduplication.

SieveComP directly addresses this need by providing an open-source, high performance validation engine tailored exclusively for SeisComP deployments. It allows operators to apply standardized, reproducible filtering rules (via TOML configurations) to ensure that discrepancies are systematically flagged before catalog publication. By providing a secure, read-only abstraction layer, SieveComP modernizes the catalog review process for human analysts and provides a safe API boundary for next-generation AI-driven monitoring agents, reducing analyst workload and improving the fidelity of published seismicity datasets.

The SieveComP workflow

SieveComP is designed with a multi-tiered workflow to accommodate a wide range of use cases. Users can interact with the package at three distinct levels of abstraction: the declarative core, the programmatic low-level, and the AI-enhanced workflows.

Declarative core

At the highest level, SieveComP operates via human-readable TOML configuration files. This allows seismologists to define complex validation pipelines without writing Python code.

- **Configuration files**: The user defines database credentials, static SQL queries to extract data from SeisComP databases, custom polygonal boundaries, and categorical, numeric, or spatial validation rules within a `.toml` file.
- **Validation engine**: The core engine reads the configuration file, translates the rules into vectorized operations, and applies them to the extracted seismic catalog. The output is a filtered catalog with flagged events, which can be exported in various formats (e.g., CSV).
- **Inspection & reporting:** The pipeline returns a `RunResult` data class containing distinct pandas DataFrames for raw data (`total_events`), logical flag results (`checks`), detected `duplicates`, and the final formatted `output`. Analysts can rapidly iterate on their validation rules by modifying the TOML file and calling the `update_from_cache()` method, which re-evaluates the rules instantly on the cached data without triggering a new database query.

Low-level workflows

For advanced users building custom applications (such as Streamlit dashboards or dynamic Jupyter notebooks), static TOML files may be too rigid. SieveComP exposes its internal architecture to allow fully programmatic workflows:

- **Dynamic Querying:** Users can intercept static SQL queries defined in the configuration and wrap them in dynamic temporal bounds to fetch specific datasets on demand.
- **In Memory Injection:** Instead of editing a text file, researchers can construct validation rules as Python dictionaries and inject them directly into the `Runner`'s active memory for instant execution.
- **Raw vectorized masking:** Users can completely bypass the orchestrator and apply SieveComP's core `NumPy` vectorized masking functions directly to their Pandas

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DataFrames. These individual masks can be linked using composite logic to evaluate highly complex, nested seismological criteria without requiring any TOML configuration.

AI-enhanced workflows

In the modern landscape of scientific computing, SieveComP can work in conjunction with Large Language Models (LLMs) and Agentic AI. Because the primary pipeline is driven by a structured, declarative TOML schema, it works as an ideal tool for AI generation.

- **LLM prompting:** Users can provide an LLM with plain-English seismological requirements (e.g., *"filter events with magnitudes greater than 4.0, depths less than 20 km, and within the specified polygonal region"*). Modern-day LLMs translate these requirements into syntactic TOML configurations.
- **Agentic AI:** SieveComP can be integrated into autonomous agents that monitor seismic catalogs, generate visualizations, and produce routine reports. By restricting AI interactions to the well-defined SieveComP read-only layer, users can upgrade the efficiency of autonomous agents while mitigating the risks of agentic AI database manipulation.

Research impact statement

The SieveComP core package is actively utilized to maintain the quality standards of the public seismic catalog published by the Red Sismológica Nacional de Colombia (RSNC) from the Servicio Geológico Colombiano (SGC) (Servicio Geológico Colombiano, 2026b). Since 2025, it has been integrated into the institution's operational workflows specifically to filter, validate, and standardize the foundational data *before* it is incorporated into the official monthly, semestral, and annual seismicity bulletins (Servicio Geológico Colombiano, 2026a).

By automating these quality assurance processes, SieveComP acts as the critical auditing layer, directly ensuring the reliability and consistency of the seismological data provided to civil authorities and the public. Consequently, any current or future research that relies on SGC-RSNC data is passively impacted by SieveComP, as the package establishes and guarantees the high-quality standard of the underlying datasets used in those scientific discoveries. # AI usage disclosure

Generative AI tools were utilized during development strictly to accelerate code implementation, generate test boilerplate, troubleshoot interfaces, and draft documentation. All AI-assisted outputs were reviewed, edited, and validated by the human author, who retained full responsibility for the framework's architecture, core logic, and all scientific and evaluative decisions.

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